

# Changin Oh

PhD Candidate  
Department of Mathematics  
Toronto Metropolitan University

c4oh@torontomu.ca  
[changinoh.github.io](https://github.com/changinoh)

---

## Education

---

### Toronto Metropolitan University

2023 – Present

*PhD in Mathematics*

*Toronto, ON*

- Supervisor: Kathleen Wilkie
- Thesis: “Reliable Mathematical Modelling in Oncology: Model Formulation, Parameter Estimation, and Data-Driven Discovery”

### York University

2021 – 2023

*MA in Mathematics*

*North York, ON*

- Supervisor: Iain Moyles
- Thesis: “Linear Spectral Unmixing Algorithms for Abundance Fraction Estimation in Spectroscopy”

### University of Western Ontario

2009 – 2017

*BSc in Mathematics*

*London, ON*

- Minor in Japanese Studies
- Dean’s Honour List for 2009 – 2011

---

## Publications

---

### Peer-Reviewed Journal Articles

1. M. Oh, **C. Oh**, T. Ghosh, and E. Tabassum, “Selective effects of backbone cyclization and disulfide bonding as global covalent constraints on the conformational ensemble of sunflower trypsin inhibitor-1,” *The Journal of Physical Chemistry B*, vol. 130, no. 28, pp. 7075–7084, Jul 2026.
2. **C. Oh** and K. P. Wilkie, “How mathematical forms of chemotherapy and radiotherapy bias model-optimized predictions: Implications for model selection,” *Mathematical Biosciences*, vol. 396, 109685, Jun 2026.
3. M. Oh, **C. Oh**, and E. Tabassum, “Covalent: Interpretable and discriminative collective variables reveal ligand-dependent switching in human cellular retinol-binding protein 2,” *Journal of Chemical Theory and Computation*, vol. 21, no. 23, pp. 12356–12370, Dec 2025.
4. M. Oh, **C. Oh**, and D. F. Weaver, “Effect of cholesterol on the structure of networked water at the surface of a model lipid membrane,” *The Journal of Physical Chemistry B*, vol. 124, no. 18, pp. 3686–3694, May 2020.
5. M. Oh, **C. Oh**, M. Gupta, and D. F. Weaver, “Decoding interfacial water orientation to predict surface charge density on a model sheet using a deep learning algorithm,” *The Journal of Physical Chemistry C*, vol. 124, no. 4, pp. 2574–2582, Jan 2020.

6. M. Oh, M. Gupta, **C. Oh**, and D. F. Weaver, “Understanding the effect of nanoconfinement on the structure of water hydrogen bond networks,” *Physical Chemistry Chemical Physics*, vol. 21, no. 47, pp. 26237–26250, Dec 2019.

## Blog Posts

1. A. Köhn-Luque, J. Nardini, **C. Oh**, L. Podina, G. Powathil, E. Rutter, J. West, and K. Wilkie, “Learning dynamics with neural networks,” *Mathematical Oncology Blog*, Mar 2025.

---

## Presentations

---

### Toroidal Search Algorithm: A Topology-Inspired Metaheuristic

- 2026 Complex Systems Day (Toronto) 2026/08
- Mathematical Biology Seminar (Toronto Metropolitan University) 2026/02

### A Gentle Introduction to Riemannian Gradient Descent Method on Smooth Manifolds

- Chemistry Research Group (Memorial University of Newfoundland) 2026/01
- Mathematical Biology Seminar (Toronto Metropolitan University) 2025/10

### Aging Effects in Cancer: An Age-Dependent Tumour-Immune Model

- 2025 Complex Systems Day (Toronto) 2025/08
- Mathematical Biology Seminar (Toronto Metropolitan University) 2024/02

### How Mathematical Forms Describing Cytotoxic Action of Chemotherapy and Radiotherapy Bias Predictions

- Workshop on Mechanistic Learning as a Combination of Machine Learning and Modelling in Mathematical Oncology (Poster – Banff International Research Station) 2025/01
- Workshop on Mathematical Modelling of Cancer Treatments, Resistance, Optimization (Poster – Fields Institute) 2024/09

### Teaching-Learning-Based Optimization with Applications in Analytic Chemistry

- Mathematical Biology Seminar (Toronto Metropolitan University) 2023/09

### Optimization Techniques for Linear Spectral Unmixing in Spectroscopy

- Mathematics & Statistics Seminar (York University) 2022/10

---

## Awards and Prizes

---

Ontario Graduate Scholarship (\$15,000 each) ..... 2024 – 2025, 2026 – 2027

Prestigious, competitive, merit-based provincial scholarship awarded to graduate students in Ontario for academic excellence.

Toronto Metropolitan Graduate Fellowship (Total: \$16,250) ..... 2023 – 2024, 2025 – 2026

Competitive, merit-based institutional fellowship for graduate students demonstrating academic excellence.

Mathematics Graduate Award (Total: \$11,650) ..... 2023 – 2026

Departmental award recognizing dedication and achievement in graduate mathematics.

Toronto Metropolitan Graduate Student Travel Fund Award (\$500) ..... 2025

Institutional support for research dissemination and active participation at academic conferences.

|                                                                                                          |             |
|----------------------------------------------------------------------------------------------------------|-------------|
| Toronto Metropolitan Graduate Scholarship (declined) (\$15,000) .....                                    | 2024 – 2025 |
| Competitive, merit-based institutional scholarship awarded to graduate students for academic excellence. |             |
| Toronto Metropolitan Graduate Development Award (\$2,550) .....                                          | 2024 – 2025 |
| Departmental award recognizing academic potential and professional success.                              |             |
| York Graduate Fellowship (\$5,403) .....                                                                 | 2021 – 2022 |
| Institutional fellowship supporting academic and research activities.                                    |             |
| Classical Studies Prize in Elementary Latin (\$150) .....                                                | 2016        |
| Awarded to the student with the highest mark in Latin 1000.                                              |             |
| Class of '49 Prize (\$150) .....                                                                         | 2010        |
| Awarded to the student with the highest mark in Calculus 1501.                                           |             |
| Western Scholarship of Excellence (\$8,000) .....                                                        | 2009        |
| Merit-based entrance scholarship recognizing outstanding high school academic achievement.               |             |

---

## Honours and Distinctions

---

|                                                                                                                                                                                                                                                                    |         |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| <b>Featured Research Article</b>                                                                                                                                                                                                                                   | 2026/05 |
| <i>Mathematical Oncology Newsletter</i>                                                                                                                                                                                                                            |         |
| <ul style="list-style-type: none"> <li>▫ The article “How mathematical forms of chemotherapy and radiotherapy bias model-optimized predictions: Implications for model selection” was featured in <i>This Week in Mathematical Oncology</i>, Issue 370.</li> </ul> |         |

---

## Research Experience

---

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                   |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| <b>Collaborating Researcher</b>                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2025/09 – Present |
| <i>Memorial University of Newfoundland</i>                                                                                                                                                                                                                                                                                                                                                                                                                             |                   |
| <i>St. John's, NL</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                   |
| <ul style="list-style-type: none"> <li>▫ Implemented and enhanced a Riemannian gradient descent algorithm for multiclass harmonic linear discriminant analysis on the Stiefel manifold.</li> <li>▫ Formulated a novel global twist angle for cyclic peptides and developed its computational implementation.</li> <li>▫ Developed an information-theoretic framework for inferring surface properties from the molecular structure of interfacial solvents.</li> </ul> |                   |
| <b>Collaborating Researcher</b>                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2019/07 – 2020/06 |
| <i>Krembil Research Institute</i>                                                                                                                                                                                                                                                                                                                                                                                                                                      |                   |
| <i>Toronto, ON</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                   |
| <ul style="list-style-type: none"> <li>▫ Conducted topological analyses of hydrogen bond networks in confined and interfacial water.</li> <li>▫ Implemented a deep learning algorithm for indirect surface characterization.</li> <li>▫ Investigated the effects of cholesterol on water structure at the surface of a model lipid membrane.</li> </ul>                                                                                                                |                   |
| <b>Volunteer Research Assistant</b>                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2015/05 – 2015/08 |
| <i>University of Western Ontario</i>                                                                                                                                                                                                                                                                                                                                                                                                                                   |                   |
| <i>London, ON</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                   |
| <ul style="list-style-type: none"> <li>▫ Investigated nanosized aqueous droplets using molecular dynamics simulations.</li> </ul>                                                                                                                                                                                                                                                                                                                                      |                   |

---

## Teaching Experience

---

### Teaching Assistant

2023/09 – 2027/04

*Toronto Metropolitan University*

*Toronto, ON*

- Led weekly labs guiding students through difficult problems and reviewing course material.
- Contributed to the development of exam marking schemes (MTH 231).

MTH 141 – Linear Algebra

MTH 207 – Calculus and Computational Methods I

MTH 231 – Modern Mathematics II

MTH 240 – Calculus II

MTH 330 – Calculus and Geometry

MTH 350 – Multivariate and Vector Calculus

MTH 501/510 – Numerical Analysis

MTH 430 – Dynamic Systems Differential Equations

MTH 719 – Applied Linear Algebra

### Guest Instructor

2026/01 – 2026/05

*Memorial University of Newfoundland*

*St. John's, NL*

- Designed and delivered intuition-focused lectures on general topology and differential geometry for a chemistry research group.

### Teaching Assistant

2021/09 – 2022/04

*York University*

*North York, ON*

- Provided one-on-one and group tutoring to students.

MATH 1013 – Applied Calculus I

MATH 1025 – Applied Linear Algebra

MATH 1300 – Differential Calculus with Applications

MATH 1507 – Mathematics for the Life Sciences

### Teaching Assistant

2013/09 – 2014/04

*University of Western Ontario*

*London, ON*

- Led group conversation tutorials to support students in improving their Japanese speaking skills.

JAPANESE 1036 – Japanese for Beginners

JAPANESE 2260 – Intermediate Japanese

---

## Educational Outreach

---

### Instructor

2026/02 – 2026/06

*Toronto Garden Church*

*North York, ON*

- Taught mathematics to students in Grades 7–10 as part of a community outreach program.

### Instructor

2025/03 – 2025/06

*Toronto Milal Church*

*Toronto, ON*

- Taught essential computer skills to elderly learners, including Microsoft Word, Paint, and introductory AI tools.

### Instructor

2021/08 – 2024/07

*Online Learning Community*

*Korea*

- Taught linear algebra, calculus, probability theory, and real analysis online to learners without strong mathematical backgrounds.

### Instructor

2015/09 – 2016/08

*London Full Gospel Church*

*London, ON*

- Taught mathematics and Japanese to secondary school students as part of a community outreach program.

**Instructor**

2014/09 – 2015/08

*Oakridge Presbyterian Church**London, ON*

- Taught mathematics to secondary school students as part of a community outreach program.

**Instructor**

2008/05 – 2009/07

*Oakville Presbyterian Church**Oakville, ON*

- Taught mathematics and Japanese to secondary school students as part of a community outreach program.

---

**Technical and Language Skills**

---

- Julia, MATLAB, Python
- Korean (native), English (advanced), Japanese (advanced, JLPT N1)